

Advanced Project Phase 3

Team Members:

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Introduction: Tuberculosis is an infectious disease that has been a significant health problem worldwide for many years. It is caused by a type of bacteria known as “Mycobacterium tuberculosis”, which primarily attacks the lungs [4]. However, Tuberculosis doesn’t just stop there, it can also spread to other parts of the body, including critical areas like the brain, spine, and kidneys [6]. Transmission occurs through inhalation of droplet nuclei containing the bacteria, expelled when an individual with active pulmonary TB coughs, sneezes, or talks [4]. Once inhaled, the bacilli reach the alveoli and are phagocytosed by alveolar macrophages. [17] The bacteria can survive and replicate within macrophages by inhibiting phagosome-lysosome fusion. The host immune response leads to the formation of granulomas, aiming to contain the infection. In some individuals, the bacteria remain dormant (latent TB infection), while in others, especially those with weakened immunity, active disease develops. [17] This makes it a dangerous disease because it can affect multiple vital organs, leading to severe health complications. The World Health Organization (WHO) highlights Tuberculosis as one of the top global health concerns, reporting over 10 million new cases every year [3]. Tragically, around 1.5 million people lose their lives to Tuberculosis annually, showing just how deadly this disease can be [3]. For instance, if someone with active Tuberculosis coughs, sneezes, or even talks, the bacteria can be released into the air. Anyone nearby, especially in crowded areas like buses, schools, or homes where people live in close quarters, could inhale the bacteria and get infected. Despite being a preventable and treatable illness, tuberculosis continues to pose a threat to global health. Efforts to combat TB require a multifaceted approach, including improving diagnostics, developing more effective vaccines and treatments, addressing drug resistance, and tackling socioeconomic determinants. Global collaboration and sustained commitment are essential to achieve the WHO's End TB Strategy targets of reducing TB incidence by 80% and deaths by 90% by 2030 compared to 2015 levels [18]. Traditional methods of Tuberculosis diagnosis, such as sputum smear microscopy and chest X rays, are often limited by the need for skilled technicians and adequate healthcare infrastructure, both of which are scarce in high-burden regions [7]. Sputum smear microscopy, one of the most widely used methods, has a low sensitivity, especially for detecting Tuberculosis in individuals with HIV or children [7]. Chest X-rays, another standard diagnostic tool, require experienced radiologists to interpret the images accurately. In recent years, artificial intelligence (AI) and machine learning (ML) have become important tools in the fight against tuberculosis. These technologies have the ability to make Tuberculosis detection more accurate and efficient, especially in areas with limited healthcare resources. Advances in deep learning, especially in types of models called convolutional neural networks (CNNs), allow computers to learn how to spot patterns in medical images, like chest X-rays, which can help identify Tuberculosis. Research has shown that AI models can sometimes be as accurate, or even more accurate, than expert radiologists in diagnosing Tuberculosis from X ray images [8]. By automating the process of analysing chest X-rays, AI can greatly ease the load on healthcare systems, providing faster diagnoses in areas that may lack access to medical professionals.

Recent Studies on the detection of Tuberculosis using Machine Learning:

1. A recent study in 2024 looked at how computer vision and machine learning can help find tuberculosis bacteria in images of human lung tissue. The study showed how AI can be used to make diagnosing Tuberculosis easier and faster, which is important for catching the disease early. This research emphasizes how AI can improve the way doctors diagnose Tuberculosis. [15]
2. A review by Guney and team in 2023 looked at different machine learning and deep learning models used to diagnose tuberculosis. The study highlighted how these technologies could help make Tuberculosis diagnosis more accurate and faster, especially in places with limited resources. [16]

Hypothesis Statements:

We hypothesize that the use of open source machine learning software applied to the Tuberculosis Chest X-rays dataset may produce useful technology for studying and detecting Tuberculosis based on the Chest X-ray images.

We hypothesize that the use of open source machine learning software applied to the Tuberculosis Chest X-rays dataset may produce useful technology for studying and detecting the accuracy of the software based on the amount of the data provided i.e. the Chest X-ray images.

Materials and Methods:

Dataset Description:

Dataset: <https://www.kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset>

This dataset includes Chest X-ray images from both healthy individuals and tuberculosis (TB) patients. There are 3,500 images of normal cases and 700 publicly accessible TB images, with an additional 2,800 TB images available for download from the NIAID TB portal after signing a data-sharing agreement. [9]

The Tuberculosis image data was gathered from several sources:

1. NLM Dataset: The National Library of Medicine (NLM) in the United States has released two lung X-ray datasets: the Montgomery and Shenzhen datasets.
2. Belarus Dataset: The Belarus Set was created for a drug-resistance study conducted by the National Institute of Allergy and Infectious Diseases and the Ministry of Health in Belarus.
3. NIAID TB Dataset: In this program portal dataset contains about 3,000 chest X-ray images from roughly 3,087 Tuberculosis positive cases. To access these images, users must agree to data-sharing terms on the NIAID website, where the images in DICOM format are available for download (<https://tbportals.niaid.nih.gov/download-data>).
4. RSNA CXR Dataset: This dataset was part of the RSNA pneumonia detection challenge and includes about 30,000 chest X-ray images. Among these, 10,000 images are of healthy cases, while the rest show abnormalities and lung opacity.

This dataset was used in the 2020 IEEE Access paper titled "Reliable Tuberculosis Detection using Chest X-ray with Deep Learning, Segmentation and Visualization. [10]

Recent Studies on the dataset:

1. An Ensemble CNN-Dempster Shafer Based Tuberculosis Detection from Chest X-rays (2023): Researchers utilized the dataset to train an ensemble of convolutional neural networks (CNNs) combined with Dempster-Shafer theory, achieving an accuracy of 94.21% in distinguishing Tuberculosis positive from normal chest X-rays. [11]
2. An Effective Identification of Tuberculosis in Chest X-rays Using Convolutional Neural Networks (2023): This work applied DenseNet121 and ResNet50 models to the dataset, employing preprocessing techniques to improve data quality. The study reported high accuracy in TB detection, with an accuracy of around 96.8%. It also achieved high precision (97.5%), sensitivity (95.2%), and an F1 score of 96.3%, demonstrating the effectiveness of these models. [12]
3. Detection and Analysis of Tuberculosis Disease from Chest X-Ray Images Using Deep Learning" (2023): Utilizing the dataset, they developed a custom deep learning model specifically designed for early diagnosis of tuberculosis. This research proposed a model for early Tuberculosis diagnosis. The custom model achieved an accuracy of about 94.2%, with precision and recall both in the range of 93.8% to 94.5%. The F1 score was approximately 94.1%, and the AUROC was 0.95, indicating strong performance. [13]
4. Reliable Tuberculosis Detection using Chest X-ray with Deep Learning, Segmentation and Visualization (2020): This study employed the dataset to develop a deep learning model for Tuberculosis detection, incorporating image preprocessing, data augmentation, and segmentation techniques to enhance diagnostic accuracy. This model achieved an accuracy of approximately 92.5%, with precision, recall, and F1 scores close to 92%, and an AUROC of 0.93. [14]

Input Variables: This dataset basically has two input variables

1. **Chest X-ray images (Normal):** The Chest X-ray images of the normal people with no evidence of Tuberculosis or any other lung related diseases.
2. **Chest X-ray images (Tuberculosis):** The Chest X-ray images of the people who were tested positive for Tuberculosis.

Sample Count: There are a total of 4200 samples available in this dataset. Out of the 4200 samples, 3500 samples are the X-rays of normal people and 700 samples are the X-rays of Tuberculosis positive people.

Target Variable: The target variable of this dataset will be the **Type or Status** of the chest X-ray image i.e. whether a particular chest X-ray image is a normal one or a tuberculosis infected one, and it comes under Binary classification.

Machine Learning:

Df-analyze:

The Machine Learning software we used for this project is df-analyze, which is an AutoML (Automated Machine Learning) tool, which aim to reduce the effort required to perform complex machine learning tasks. The tool is command-line based, meaning users interact with it through text commands in a terminal or command prompt. Currently, it is distributed as Python scripts, which come with a set of required dependencies that must be installed to ensure smooth functioning.[2]

One of the standout features of "df-analyze" is its ability to run multiple machine learning models on the same dataset at the same time. This feature allows users to compare the performance of different models without manually running each one. Once the models are executed, the tool generates an output that includes performance metrics, selected features, and other relevant information. This output can be analyzed to determine which model is best suited for a given task.

The results generated by "df-analyze" are stored in a structured format for easy analysis. Users receive detailed information about the performance of each model, along with the selected features and other important insights. These outputs help users identify the best-performing model for their specific task and understand the key factors influencing the predictions.

X-Vision helper: The X-Vision Helper is a command line tool makes it easy to fine-tune the ResNet-50 deep learning model for image classification tasks. It simplifies the process by managing things like loading data, training the model. It also lets users extract embeddings from the model. The embeddings produced by the df-analyze can be used with df-analyze to test different models, such as logistic regression or SVMs. [1]

Statistical Analysis:

Df-analyze tests several machine learning models at the same time and compares how well they perform. Some of the models it checks are Logistic Regression, Random Forest, Stochastic Gradient Descent, LightGBM, and more. For each model, df-analyze tries different ways of selecting the best features and using embeddings to find the best setup.

The tool measures performance using many metrics like accuracy (how often the model is right), AUROC (how well it separates classes), F1 score (a balance between precision and recall), sensitivity (how well it finds positive cases), specificity (how well it finds negative cases), and balanced accuracy (a fair measure for uneven datasets). These metrics help df-analyze pick a model that works well overall and handles imbalanced data effectively.

To validate model performance, df-analyze employs holdout set evaluation and cross-validation techniques. The holdout set provides a snapshot of how well the models generalize to unseen data. Cross-validation further ensures that the models are not overly dependent on specific training samples, improving the reliability of the analysis.

Though there are a lot of statistical measures, we are relying on the following measures of df-analyze:

AUROC (Area Under the Receiver Operating Characteristic Curve):

This metric measures how well a model can tell the difference between two classes, like positive cases (people with tuberculosis) and negative cases (people without tuberculosis). The AUROC shows how good the model is at ranking predictions by their probability of belonging to a class. A high AUROC score means the model performs well over a range of decision thresholds, consistently distinguishing between the two classes. For example, a model with a high AUROC can reliably identify both true positives and true negatives, regardless of where the threshold is set.

Balanced Accuracy:

Balanced accuracy is a helpful metric when working with imbalanced datasets, where one class has significantly more samples than the other. It calculates the average of two important measures: sensitivity (how well the model identifies positive cases) and specificity (how well the model identifies negative cases). This way, balanced accuracy gives equal importance to both classes, ensuring a fair evaluation of the model's performance even if the dataset is not evenly distributed. For example, if there are far fewer tuberculosis cases than normal cases, balanced accuracy ensures the model is judged fairly by considering both types of predictions equally.

F1 Score:

The F1 score is a measure that combines precision (how many of the predicted positive cases are actually correct) and recall (how many of the actual positive cases the model correctly identifies). It does this by calculating the harmonic mean of these two values, giving a single score that balances both aspects. The F1 score is especially useful when the dataset is imbalanced, as it accounts for both false positives and false negatives. For instance, in a tuberculosis detection task, a high F1 score means the model is not only identifying most TB cases but also making very few mistakes in labeling healthy cases as TB-positive.

Hypothesis Statement 1: We hypothesized that the use of open source machine learning software applied to the Tuberculosis Chest X-rays dataset may produce useful technology for studying and detecting Tuberculosis based on the Chest X-ray images.

Results: In this experiment we ran a total of 4200 chest X-ray images through df-analyze and got good results in a single run.

Holdout Set Performance

The evaluation of the models on the holdout set showed that Logistic Regression (LR) was the best-performing model. It achieved an accuracy of 99.7%, an AUROC of 1.000, a balanced accuracy of 99.1%, and an F1 score of 99.5%. Another strong model was Stochastic Gradient Descent (SGD), which performed slightly lower than LR, with an accuracy of 99.6%, an AUROC of 0.998, a balanced accuracy of 98.7%, and an F1 score of 99.2%. Both models performed well in distinguishing between tuberculosis and normal chest X-rays, showing high sensitivity and specificity.

5-Fold Performance on Holdout Set

The models were also validated using 5-fold cross-validation, and the results were consistent with the holdout set performance. Logistic Regression (LR) again emerged as the best model, achieving an accuracy of 99.6%, an AUROC of 1.000, a balanced accuracy of 98.9%, and an F1 score of 99.3%. Stochastic Gradient Descent (SGD) followed closely, with an accuracy of 99.4%, an AUROC of 0.994, a balanced accuracy of 98.4%, and an F1 score of 98.9%. Other models, like LightGBM (LGBM) and Random Forest (RF), also performed well but did not match the accuracy and AUROC of LR and SGD.

Holdout Set Performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lr	none	none	0.997	1	0.991	0.995	1	0.997	0.991	0.983
lr	assoc	none	0.997	1	0.99	0.994	1	0.996	0.99	0.98
lr	pred	none	0.996	1	0.988	0.993	1	0.995	0.988	0.977
lr	embed_linear	linear	0.996	1	0.988	0.993	1	0.995	0.988	0.977
sgd	assoc	none	0.996	0.998	0.987	0.992	1	0.995	0.987	0.974
sgd	pred	none	0.995	0.995	0.989	0.991	0.991	0.996	0.989	0.98
sgd	none	none	0.995	0.997	0.988	0.991	0.994	0.995	0.988	0.977
sgd	embed_lgbm	lgbm	0.993	0.999	0.983	0.987	0.988	0.994	0.983	0.968
lr	embed_lgbm	lgbm	0.993	1	0.983	0.987	0.988	0.994	0.983	0.968
lgbm	embed_lgbm	lgbm	0.992	1	0.976	0.985	1	0.99	0.976	0.951
sgd	embed_linear	linear	0.992	0.999	0.979	0.985	0.991	0.992	0.979	0.96
lgbm	pred	none	0.99	0.999	0.971	0.981	0.997	0.989	0.971	0.942
lgbm	assoc	none	0.989	0.999	0.967	0.98	1	0.987	0.967	0.934
lgbm	none	none	0.989	1	0.967	0.98	1	0.987	0.967	0.934
knn	embed_lgbm	lgbm	0.989	0.99	0.967	0.98	1	0.987	0.967	0.934
lgbm	embed_linear	linear	0.989	1	0.965	0.979	1	0.986	0.965	0.931
knn	pred	none	0.982	0.982	0.948	0.966	0.994	0.98	0.948	0.896
knn	none	none	0.98	0.984	0.943	0.962	0.99	0.978	0.943	0.888
knn	embed_linear	linear	0.98	0.984	0.943	0.962	0.99	0.978	0.943	0.888
knn	assoc	none	0.98	0.984	0.943	0.962	0.99	0.978	0.943	0.888
rf	embed_linear	linear	0.979	0.993	0.944	0.96	0.975	0.979	0.944	0.893
rf	embed_lgbm	lgbm	0.979	0.995	0.941	0.959	0.984	0.978	0.941	0.885
rf	none	none	0.978	0.995	0.943	0.959	0.975	0.979	0.943	0.89
rf	pred	none	0.976	0.994	0.939	0.954	0.965	0.978	0.939	0.885
rf	assoc	none	0.976	0.993	0.936	0.954	0.974	0.976	0.936	0.876
lr	wrap	none	0.956	0.966	0.887	0.915	0.941	0.959	0.887	0.784
sgd	wrap	none	0.956	0.965	0.888	0.914	0.935	0.959	0.888	0.787
knn	wrap	none	0.955	0.934	0.883	0.912	0.941	0.957	0.883	0.775
lgbm	wrap	none	0.954	0.964	0.885	0.911	0.931	0.958	0.885	0.781
rf	wrap	none	0.954	0.941	0.892	0.913	0.914	0.961	0.892	0.798
dummy	embed_lgbm	lgbm	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	pred	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	none	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0

dummy	embed_linear	linear	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	assoc	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0

5 – fold performance on Holdout Set

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lr	assoc	none	0.996	1	0.989	0.993	1	0.995	0.989	0.977
lr	none	none	0.996	1	0.987	0.992	1	0.995	0.987	0.974
lr	embed_linear	linear	0.995	1	0.987	0.991	0.997	0.995	0.987	0.974
lr	embed_lgbm	lgbm	0.994	0.999	0.985	0.989	0.994	0.994	0.985	0.971
sgd	pred	none	0.994	0.994	0.984	0.989	0.994	0.994	0.984	0.968
lr	pred	none	0.994	1	0.982	0.989	0.997	0.993	0.982	0.966
sgd	assoc	none	0.994	0.998	0.984	0.989	0.994	0.994	0.984	0.968
sgd	none	none	0.993	0.995	0.985	0.988	0.988	0.994	0.985	0.971
sgd	embed_linear	linear	0.992	0.999	0.98	0.986	0.991	0.993	0.98	0.963
sgd	embed_lgbm	lgbm	0.991	0.999	0.981	0.984	0.982	0.993	0.981	0.966
lgbm	embed_lgbm	lgbm	0.99	0.999	0.971	0.981	0.997	0.989	0.971	0.943
lgbm	none	none	0.989	0.999	0.969	0.979	0.994	0.988	0.969	0.94
lgbm	assoc	none	0.989	0.999	0.968	0.979	0.994	0.988	0.968	0.937
lgbm	embed_linear	linear	0.989	0.999	0.968	0.978	0.994	0.988	0.968	0.937
lgbm	pred	none	0.988	0.999	0.968	0.978	0.991	0.988	0.968	0.937
knn	embed_lgbm	lgbm	0.984	0.986	0.954	0.97	0.997	0.982	0.954	0.908
knn	pred	none	0.978	0.977	0.939	0.958	0.987	0.977	0.939	0.879
knn	none	none	0.976	0.977	0.932	0.954	0.991	0.974	0.932	0.865
knn	embed_linear	linear	0.976	0.977	0.932	0.954	0.991	0.974	0.932	0.865
knn	assoc	none	0.976	0.977	0.932	0.954	0.991	0.974	0.932	0.865
rf	none	none	0.975	0.995	0.937	0.952	0.969	0.977	0.937	0.88
rf	embed_linear	linear	0.972	0.992	0.927	0.945	0.967	0.973	0.927	0.859
rf	pred	none	0.972	0.992	0.931	0.946	0.956	0.975	0.931	0.871
rf	assoc	none	0.969	0.993	0.927	0.941	0.948	0.973	0.927	0.862
rf	embed_lgbm	lgbm	0.969	0.993	0.923	0.941	0.958	0.972	0.923	0.854
knn	wrap	none	0.954	0.921	0.89	0.907	0.919	0.961	0.89	0.793
rf	wrap	none	0.954	0.945	0.894	0.909	0.911	0.963	0.894	0.805
lr	wrap	none	0.952	0.967	0.879	0.901	0.932	0.957	0.879	0.77
lgbm	wrap	none	0.951	0.965	0.88	0.898	0.922	0.958	0.88	0.773
sgd	wrap	none	0.951	0.965	0.883	0.9	0.91	0.959	0.883	0.782
dummy	embed_lgbm	lgbm	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	pred	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	none	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	embed_linear	linear	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	assoc	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0

Hypothesis Statement 2: We hypothesized that the use of open source machine learning software applied to the Tuberculosis Chest X-rays dataset may produce useful technology for studying and detecting the accuracy of the software based on the amount of the data provided i.e. the Chest X-ray images.

Results:

- 1. 50% of the data:** For this experiment we manually down sampled the images from both the classes to 50%, extracted the embeddings from df-embed and ran the embeddings through df-analyze.

The evaluation of the machine learning models on the holdout set of the 50% data shows that most models performed very well, with some achieved excellent accuracy and balanced results. Logistic Regression (LR) stood out as one of the best models, with an accuracy of 99.4%. Another strong performer was the Stochastic Gradient Descent (SGD) model, which achieved similar accuracy and showed reliable results. The Light Gradient Boosting Machine (LGBM) also performed well, with accuracy around 98.9% and slightly lower accuracy compared to LR and SGD. K-Nearest Neighbours (KNN) followed closely, showing an accuracy of 98.4%. However, models that used simpler selection methods, like the "wrap" approach, performed less effectively, with accuracy scores ranging from 83.1% to 85.2%.

When looking at the 5-fold cross-validation results, Logistic Regression showed the best results, achieving an accuracy of 99.5% and a well-balanced performance across all metrics. The SGD models also maintained their reliability with accuracy close to 99.2%, while LGBM models achieved accuracy between 98.6% and 98.7%.

Holdout Set Performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lr	assoc	none	0.994	0.998	0.983	0.989	1	0.993	0.983	0.965
sgd	embed_linear	linear	0.994	0.997	0.983	0.989	1	0.993	0.983	0.965
lr	none	none	0.994	0.998	0.983	0.989	1	0.993	0.983	0.965
lr	embed_linear	linear	0.994	0.998	0.983	0.989	1	0.993	0.983	0.965
sgd	assoc	none	0.993	0.997	0.982	0.988	0.994	0.993	0.982	0.965
sgd	pred	none	0.993	0.985	0.982	0.988	0.994	0.993	0.982	0.965
sgd	none	none	0.993	0.997	0.982	0.988	0.994	0.993	0.982	0.965
lr	pred	none	0.993	0.998	0.982	0.988	0.994	0.993	0.982	0.965
lr	embed_lgbm	lgbm	0.993	0.997	0.982	0.988	0.994	0.993	0.982	0.965
lgbm	embed_linear	linear	0.989	0.997	0.965	0.979	1	0.986	0.965	0.931
lgbm	embed_lgbm	lgbm	0.989	0.997	0.968	0.979	0.994	0.988	0.968	0.936
sgd	embed_lgbm	lgbm	0.989	0.997	0.975	0.979	0.976	0.991	0.975	0.954
lgbm	assoc	none	0.988	0.996	0.962	0.977	1	0.985	0.962	0.925
lgbm	none	none	0.986	0.997	0.961	0.973	0.988	0.985	0.961	0.925
lgbm	pred	none	0.984	0.998	0.956	0.97	0.988	0.983	0.956	0.913
knn	embed_lgbm	lgbm	0.984	0.988	0.951	0.969	1	0.981	0.951	0.902
knn	none	none	0.983	0.985	0.948	0.967	1	0.98	0.948	0.896
knn	embed_linear	linear	0.983	0.985	0.948	0.967	1	0.98	0.948	0.896
knn	assoc	none	0.983	0.985	0.948	0.967	1	0.98	0.948	0.896
knn	pred	none	0.982	0.988	0.945	0.966	1	0.979	0.945	0.89

rf	embed_linear	linear	0.98	0.992	0.956	0.963	0.958	0.984	0.956	0.919
rf	pred	none	0.976	0.988	0.942	0.955	0.963	0.979	0.942	0.89
rf	embed_lgbm	lgbm	0.976	0.986	0.942	0.955	0.963	0.979	0.942	0.89
rf	none	none	0.975	0.989	0.941	0.954	0.957	0.979	0.941	0.89
rf	assoc	none	0.975	0.992	0.946	0.954	0.945	0.981	0.946	0.902
lr	wrap	none	0.852	0.703	0.559	0.566	0.875	0.852	0.559	0.121
sgd	wrap	none	0.851	0.703	0.554	0.557	0.905	0.85	0.554	0.11
lgbm	wrap	none	0.843	0.664	0.565	0.576	0.591	0.854	0.565	0.15
dummy	embed_lgbm	lgbm	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
rf	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
knn	wrap	none	0.835	0.616	0.5	0.455	nan	0.835	0.5	0
dummy	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	pred	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	none	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	embed_linear	linear	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	assoc	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0

5 – fold performance on Holdout Set

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lr	embed_lgbm	lgbm	0.995	0.997	0.985	0.991	1	0.994	0.985	0.971
lr	pred	none	0.993	0.997	0.982	0.988	0.994	0.993	0.982	0.965
lr	none	none	0.992	0.997	0.979	0.986	0.994	0.992	0.979	0.959
lr	embed_linear	linear	0.992	0.997	0.979	0.986	0.994	0.992	0.979	0.959
sgd	pred	none	0.992	0.989	0.986	0.986	0.977	0.995	0.986	0.977
sgd	none	none	0.992	0.994	0.981	0.986	0.988	0.993	0.981	0.965
lr	assoc	none	0.991	0.997	0.976	0.984	0.994	0.991	0.976	0.954
sgd	assoc	none	0.991	0.997	0.976	0.984	0.994	0.991	0.976	0.954
sgd	embed_linear	linear	0.991	0.997	0.976	0.984	0.994	0.991	0.976	0.954
sgd	embed_lgbm	lgbm	0.99	0.996	0.98	0.982	0.977	0.993	0.98	0.965
lgbm	assoc	none	0.988	0.995	0.962	0.977	1	0.985	0.962	0.925
lgbm	none	none	0.987	0.995	0.962	0.975	0.994	0.985	0.962	0.925
lgbm	embed_lgbm	lgbm	0.987	0.996	0.959	0.975	1	0.984	0.959	0.919
lgbm	embed_linear	linear	0.986	0.996	0.959	0.973	0.994	0.984	0.959	0.919
knn	embed_lgbm	lgbm	0.985	0.982	0.954	0.971	1	0.982	0.954	0.908
lgbm	pred	none	0.985	0.996	0.958	0.971	0.988	0.984	0.958	0.919
knn	pred	none	0.978	0.979	0.936	0.958	0.994	0.976	0.936	0.873
knn	none	none	0.978	0.973	0.933	0.958	1	0.974	0.933	0.867
knn	embed_linear	linear	0.978	0.973	0.933	0.958	1	0.974	0.933	0.867
knn	assoc	none	0.978	0.973	0.933	0.958	1	0.974	0.933	0.867
rf	assoc	none	0.976	0.991	0.942	0.955	0.963	0.979	0.942	0.89
rf	embed_linear	linear	0.974	0.992	0.938	0.952	0.958	0.977	0.938	0.884
rf	none	none	0.974	0.989	0.931	0.951	0.974	0.974	0.931	0.867
rf	pred	none	0.97	0.987	0.917	0.943	0.979	0.969	0.917	0.838
rf	embed_lgbm	lgbm	0.968	0.987	0.916	0.938	0.96	0.969	0.916	0.838
sgd	wrap	none	0.852	0.704	0.561	0.569	0.881	0.852	0.561	0.127

lr	wrap	none	0.85	0.704	0.555	0.56	0.874	0.851	0.555	0.115
dummy	embed_lgbm	lgbm	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
rf	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
knn	wrap	none	0.835	0.566	0.5	0.455	nan	0.835	0.5	0
dummy	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	pred	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	none	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	embed_linear	linear	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	assoc	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
lgbm	wrap	none	0.831	0.699	0.519	0.486	0.409	0.841	0.519	0.053

2. 75% of the data

Results: For this experiment we manually down sampled the images from both the classes to 75%, extracted the embeddings from df-embed and ran the embeddings through df-analyze

The results for the models using 75% of the dataset show that most of them performed very well. The Stochastic Gradient Descent (SGD) model with special feature selection performed the best on the holdout set, with an accuracy of 99.4%, a high AUROC score of 98.7%, and a balanced accuracy of 98.7%. Logistic Regression (LR) models were close behind, with accuracies ranging from 99.2% to 99.3% and balanced accuracy between 97.8% and 98.0%. These models also showed good results in terms of sensitivity and specificity.

The Light Gradient Boosting Machine (LGBM) and K-Nearest Neighbours (KNN) models also performed well. LGBM had an accuracy of around 98.7%, while KNN models achieved about 98.3%. These models were not as strong as SGD and LR but still showed good overall performance.

The 5-fold cross-validation results were similar to the holdout set. The SGD models again performed the best, with an accuracy of 99.7%, followed by LR models with advanced features at 99.6%. These models also had high balanced accuracy, reaching 99.0% or close to it. LGBM and KNN models also continued to perform well, with accuracies between 98.0% and 98.7%.

In summary, SGD and LR models with advanced feature selection were the most accurate and reliable for this dataset. These models are great choices for tasks that need high precision and dependability.

Holdout Set Performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
sgd	assoc	none	0.994	0.987	0.987	0.988	0.984	0.995	0.987	0.977
lr	pred	none	0.993	0.998	0.979	0.987	1	0.992	0.979	0.958
lr	embed_lgbm	lgbm	0.993	0.998	0.98	0.987	0.996	0.992	0.98	0.962
lr	assoc	none	0.992	0.998	0.978	0.986	0.996	0.992	0.978	0.958
sgd	pred	none	0.992	0.996	0.978	0.986	0.996	0.992	0.978	0.958
sgd	none	none	0.992	0.996	0.978	0.986	0.996	0.992	0.978	0.958

sgd	embed_lgbm	lgbm	0.992	0.998	0.978	0.986	0.996	0.992	0.978	0.958
lr	none	none	0.992	0.998	0.978	0.986	0.996	0.992	0.978	0.958
lr	embed_linear	linear	0.992	0.998	0.978	0.986	0.996	0.992	0.978	0.958
sgd	embed_linear	linear	0.991	0.998	0.976	0.984	0.992	0.991	0.976	0.954
lgbm	none	none	0.988	0.997	0.967	0.978	0.992	0.987	0.967	0.935
knn	embed_lgbm	lgbm	0.987	0.99	0.962	0.976	1	0.985	0.962	0.923
lgbm	pred	none	0.987	0.997	0.965	0.976	0.992	0.986	0.965	0.931
lgbm	embed_linear	linear	0.987	0.998	0.963	0.975	0.992	0.986	0.963	0.927
lgbm	embed_lgbm	lgbm	0.987	0.997	0.963	0.975	0.992	0.986	0.963	0.927
lgbm	assoc	none	0.985	0.997	0.957	0.973	0.996	0.983	0.957	0.915
knn	none	none	0.983	0.982	0.948	0.968	1	0.98	0.948	0.896
knn	embed_linear	linear	0.983	0.982	0.948	0.968	1	0.98	0.948	0.896
knn	assoc	none	0.983	0.982	0.948	0.968	1	0.98	0.948	0.896
knn	pred	none	0.981	0.984	0.945	0.964	0.991	0.979	0.945	0.892
rf	none	none	0.978	0.992	0.947	0.959	0.963	0.98	0.947	0.9
rf	assoc	none	0.978	0.991	0.947	0.959	0.963	0.98	0.947	0.9
rf	pred	none	0.977	0.99	0.945	0.957	0.963	0.98	0.945	0.896
rf	embed_lgbm	lgbm	0.976	0.991	0.944	0.956	0.959	0.98	0.944	0.896
rf	embed_linear	linear	0.975	0.989	0.942	0.954	0.955	0.979	0.942	0.892
lr	wrap	none	0.962	0.971	0.905	0.927	0.942	0.965	0.905	0.819
sgd	wrap	none	0.96	0.963	0.904	0.924	0.93	0.965	0.904	0.819
knn	wrap	none	0.957	0.922	0.88	0.914	0.966	0.955	0.88	0.765
lgbm	wrap	none	0.955	0.966	0.905	0.917	0.893	0.967	0.905	0.831
rf	wrap	none	0.953	0.962	0.896	0.911	0.894	0.963	0.896	0.812
dummy	embed_lgbm	lgbm	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	pred	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	none	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	embed_linear	linear	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	assoc	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0

5 – fold performance on Holdout Set

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
sgd	none	none	0.997	0.998	0.99	0.994	1	0.996	0.99	0.981
lr	embed_lgbm	lgbm	0.996	0.999	0.99	0.993	0.996	0.996	0.99	0.981
sgd	pred	none	0.996	0.998	0.99	0.993	0.996	0.996	0.99	0.981
sgd	embed_linear	linear	0.996	0.998	0.99	0.993	0.996	0.996	0.99	0.981
lr	pred	none	0.996	0.999	0.988	0.993	1	0.995	0.988	0.977
lr	none	none	0.996	1	0.988	0.993	1	0.995	0.988	0.977
lr	embed_linear	linear	0.996	1	0.988	0.993	1	0.995	0.988	0.977
lr	assoc	none	0.996	0.999	0.987	0.992	1	0.995	0.987	0.973
sgd	embed_lgbm	lgbm	0.996	0.998	0.987	0.992	1	0.995	0.987	0.973
sgd	assoc	none	0.995	0.991	0.991	0.991	0.985	0.997	0.991	0.985
lgbm	embed_linear	linear	0.989	0.997	0.967	0.979	0.996	0.987	0.967	0.935
knn	embed_lgbm	lgbm	0.987	0.982	0.962	0.976	1	0.985	0.962	0.923

lgbm	none	none	0.987	0.997	0.966	0.976	0.988	0.987	0.966	0.935
lgbm	embed_lgbm	lgbm	0.987	0.997	0.961	0.975	0.996	0.985	0.961	0.923
lgbm	assoc	none	0.987	0.998	0.961	0.975	0.996	0.985	0.961	0.923
lgbm	pred	none	0.985	0.997	0.957	0.972	0.996	0.984	0.957	0.915
rf	none	none	0.98	0.993	0.945	0.963	0.987	0.979	0.945	0.892
rf	embed_lgbm	lgbm	0.98	0.992	0.945	0.963	0.987	0.979	0.945	0.892
rf	pred	none	0.98	0.994	0.948	0.962	0.975	0.981	0.948	0.9
knn	pred	none	0.98	0.978	0.938	0.961	1	0.976	0.938	0.877
rf	embed_linear	linear	0.98	0.992	0.951	0.962	0.968	0.982	0.951	0.908
rf	assoc	none	0.98	0.99	0.943	0.961	0.987	0.978	0.943	0.888
knn	none	none	0.978	0.978	0.935	0.959	1	0.975	0.935	0.869
knn	embed_linear	linear	0.978	0.978	0.935	0.959	1	0.975	0.935	0.869
knn	assoc	none	0.978	0.978	0.935	0.959	1	0.975	0.935	0.869
lr	wrap	none	0.963	0.97	0.899	0.928	0.969	0.962	0.899	0.804
sgd	wrap	none	0.962	0.962	0.903	0.928	0.954	0.964	0.903	0.815
rf	wrap	none	0.959	0.962	0.903	0.923	0.928	0.965	0.903	0.819
lgbm	wrap	none	0.959	0.956	0.905	0.923	0.927	0.966	0.905	0.823
knn	wrap	none	0.955	0.901	0.878	0.912	0.963	0.955	0.878	0.762
dummy	embed_lgbm	lgbm	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	pred	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	none	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	embed_linear	linear	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	assoc	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0

Comparison of Results Across Different Dataset Sizes

The performance of machine learning models across the entire dataset, 75% of the dataset, and 50% of the dataset is summarized below, highlighting the trends in accuracy, AUROC, balanced accuracy, and other metrics.

Entire Dataset

The results from the entire dataset demonstrate consistently excellent performance across all models. Logistic Regression (LR), Stochastic Gradient Descent (SGD), and Light Gradient Boosting Machine (LGBM) achieve near-perfect scores, with accuracy, balanced accuracy, and AUROC metrics all reaching or very close to 1.000. The 5-fold cross-validation results showed similar trends, with top models maintaining balanced accuracies above 0.99, highlighting their robustness.

75% Dataset

Using 75% of the data, the models still perform remarkably well but exhibit a slight reduction in accuracy and balanced accuracy compared to the full dataset. SGD and LR models retain leading positions, with accuracies reaching up to 0.997 and balanced accuracies around 0.990. LGBM models, while slightly behind, still achieve high performance with accuracies up to 0.992. In 5-fold cross-validation, the trends remain similar, with balanced accuracy marginally

dropping for some models but still maintaining strong performance across metrics. This demonstrates that reducing the dataset to 75% has minimal impact on the best-performing models.

50% Dataset

With only 50% of the data, the performance shows more noticeable variation among the models. While SGD and LR continue to excel, achieving accuracies up to 0.994 and balanced accuracies around 0.983, some models, such as K-Nearest Neighbours (KNN) and LGBM, experience a slight decline in metrics, with accuracies dropping to 0.987–0.990. The 5-fold cross-validation results also reflect this trend, with balanced accuracies slightly lower compared to larger dataset sizes. Despite the reduced data, the best-performing models demonstrate resilience, though simpler feature selection methods result in greater performance gaps.

- **Impact of Dataset Size:** The overall performance decreases slightly as the dataset size decreases. This is more noticeable for models with simpler feature selection methods or less robust algorithms like KNN and Random Forest (RF).
- **Consistency of Top Models:** LR and SGD consistently rank as top performers across all dataset sizes, maintaining high accuracy and balanced accuracy due to their adaptability and effective feature selection strategies.
- **Robustness in Metrics:** Even with 50% of the data, the best models maintain strong results, demonstrating their ability to generalize effectively despite reduced training data.

Fine tuning a lightweight deep learner: For this task, we used X-Vision helper [1] a pre-trained ResNet-50 model, a deep learning architecture known for its lightweight design and ability to learn detailed features from data. After training, the embeddings generated by ResNet-50 were used with df-analyze to compare and understand their results better. To ensure reliable results, we've split the dataset into three parts: Training, Validation, and Testing.

Dataset Preparation

The dataset was divided into three subsets:

1. The **Training set** was used to train the model how to make predictions.
2. The **Validation set** helped tune the model's settings and prevent overfitting.
3. The **Testing set** was saved for the final evaluation to see how well the model worked on new data.

For fine tuning the ResNet-50, we have split our dataset in three different instances and got the results from the df-analyze.

1. Training 80% Validation 10% and Testing 10%

Number of Epochs: 10

Learning Rate: 0.001

Test Accuracy: 82%

Results:

Holdout Set Performance

The performance of machine learning models using ResNet-50 embeddings on the holdout set demonstrates strong results across most algorithms. Logistic Regression (LR) achieved the highest accuracy of 98.3%, along with an AUROC of 99.7% and balanced accuracy of 96.3%. LR models consistently showed reliable sensitivity (96.3%) and specificity (93.2%).

Light Gradient Boosting Machine (LGBM) also performed well, achieving an accuracy of 98.1%, AUROC of 99.7%, and balanced accuracy of 95.7%. K-Nearest Neighbours (KNN) and Stochastic Gradient Descent (SGD) models showed slightly lower but competitive accuracies, ranging from 97.0% to 97.3%, with balanced accuracies around 91.2% to 95.5%.

5-Fold Performance on Holdout Set

Logistic Regression with LGBM embeddings remained the top-performing model, achieving an accuracy of 98.0%, AUROC of 99.7%, and balanced accuracy of 95.5%. LGBM models using various feature selection strategies achieved accuracies ranging from 97.3% to 97.7%, with AUROC scores exceeding 99.4% and balanced accuracies around 93.4% to 95.5%. SGD models showed balanced accuracies between 87.4% and 94.5%, indicating variability in performance based on feature selection methods.

Holdout Set Performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
sgd	none	none	0.997	0.998	0.99	0.994	1	0.996	0.99	0.981
lr	embed_lgbm	lgbm	0.996	0.999	0.99	0.993	0.996	0.996	0.99	0.981
sgd	pred	none	0.996	0.998	0.99	0.993	0.996	0.996	0.99	0.981
sgd	embed_linear	linear	0.996	0.998	0.99	0.993	0.996	0.996	0.99	0.981
lr	pred	none	0.996	0.999	0.988	0.993	1	0.995	0.988	0.977
lr	none	none	0.996	1	0.988	0.993	1	0.995	0.988	0.977
lr	embed_linear	linear	0.996	1	0.988	0.993	1	0.995	0.988	0.977
lr	assoc	none	0.996	0.999	0.987	0.992	1	0.995	0.987	0.973
sgd	embed_lgbm	lgbm	0.996	0.998	0.987	0.992	1	0.995	0.987	0.973
sgd	assoc	none	0.995	0.991	0.991	0.991	0.985	0.997	0.991	0.985
lgbm	embed_linear	linear	0.989	0.997	0.967	0.979	0.996	0.987	0.967	0.935
knn	embed_lgbm	lgbm	0.987	0.982	0.962	0.976	1	0.985	0.962	0.923
lgbm	none	none	0.987	0.997	0.966	0.976	0.988	0.987	0.966	0.935
lgbm	embed_lgbm	lgbm	0.987	0.997	0.961	0.975	0.996	0.985	0.961	0.923
lgbm	assoc	none	0.987	0.998	0.961	0.975	0.996	0.985	0.961	0.923
lgbm	pred	none	0.985	0.997	0.957	0.972	0.996	0.984	0.957	0.915
rf	none	none	0.98	0.993	0.945	0.963	0.987	0.979	0.945	0.892
rf	embed_lgbm	lgbm	0.98	0.992	0.945	0.963	0.987	0.979	0.945	0.892
rf	pred	none	0.98	0.994	0.948	0.962	0.975	0.981	0.948	0.9
knn	pred	none	0.98	0.978	0.938	0.961	1	0.976	0.938	0.877
rf	embed_linear	linear	0.98	0.992	0.951	0.962	0.968	0.982	0.951	0.908
rf	assoc	none	0.98	0.99	0.943	0.961	0.987	0.978	0.943	0.888

knn	none	none	0.978	0.978	0.935	0.959	1	0.975	0.935	0.869
knn	embed_linear	linear	0.978	0.978	0.935	0.959	1	0.975	0.935	0.869
knn	assoc	none	0.978	0.978	0.935	0.959	1	0.975	0.935	0.869
lr	wrap	none	0.963	0.97	0.899	0.928	0.969	0.962	0.899	0.804
sgd	wrap	none	0.962	0.962	0.903	0.928	0.954	0.964	0.903	0.815
rf	wrap	none	0.959	0.962	0.903	0.923	0.928	0.965	0.903	0.819
lgbm	wrap	none	0.959	0.956	0.905	0.923	0.927	0.966	0.905	0.823
knn	wrap	none	0.955	0.901	0.878	0.912	0.963	0.955	0.878	0.762
dummy	embed_lgbm	lgbm	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	wrap	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	pred	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	none	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	embed_linear	linear	0.835	0.5	0.5	0.455	nan	0.835	0.5	0
dummy	assoc	none	0.835	0.5	0.5	0.455	nan	0.835	0.5	0

5 – fold performance on Holdout Set

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lr	embed_lgbm	lgbm	0.98	0.997	0.955	0.963	0.96	0.984	0.955	0.918
lgbm	embed_lgbm	lgbm	0.977	0.995	0.942	0.957	0.969	0.978	0.942	0.889
lgbm	assoc	none	0.974	0.994	0.939	0.952	0.958	0.978	0.939	0.886
lgbm	none	none	0.974	0.995	0.934	0.951	0.966	0.976	0.934	0.875
lgbm	embed_linear	linear	0.973	0.994	0.935	0.95	0.959	0.976	0.935	0.879
lr	none	none	0.972	0.992	0.929	0.947	0.965	0.973	0.929	0.864
lr	embed_linear	linear	0.972	0.992	0.929	0.947	0.965	0.973	0.929	0.864
lr	assoc	none	0.972	0.992	0.929	0.947	0.965	0.973	0.929	0.864
sgd	embed_lgbm	lgbm	0.971	0.946	0.945	0.947	0.919	0.982	0.945	0.907
lr	pred	none	0.97	0.992	0.922	0.943	0.969	0.971	0.922	0.85
knn	embed_lgbm	lgbm	0.968	0.973	0.914	0.939	0.975	0.967	0.914	0.832
knn	embed_linear	linear	0.968	0.964	0.911	0.938	0.98	0.966	0.911	0.825
knn	assoc	none	0.968	0.964	0.911	0.938	0.98	0.966	0.911	0.825
knn	none	none	0.968	0.964	0.911	0.938	0.98	0.966	0.911	0.825
lgbm	pred	none	0.964	0.987	0.916	0.933	0.939	0.969	0.916	0.843
knn	pred	none	0.964	0.96	0.899	0.93	0.984	0.962	0.899	0.8
rf	embed_linear	linear	0.964	0.991	0.928	0.934	0.907	0.975	0.928	0.875
rf	embed_lgbm	lgbm	0.961	0.991	0.923	0.929	0.905	0.973	0.923	0.864
rf	none	none	0.956	0.988	0.921	0.921	0.873	0.974	0.921	0.868
rf	assoc	none	0.956	0.99	0.915	0.92	0.885	0.971	0.915	0.854
sgd	none	none	0.951	0.965	0.874	0.904	0.942	0.953	0.874	0.757
sgd	pred	none	0.951	0.963	0.868	0.902	0.952	0.951	0.868	0.743
sgd	embed_linear	linear	0.947	0.964	0.875	0.898	0.906	0.955	0.875	0.768
rf	pred	none	0.941	0.976	0.878	0.89	0.862	0.957	0.878	0.782
sgd	assoc	none	0.935	0.891	0.891	0.885	0.8	0.965	0.891	0.825
dummy	assoc	none	0.833	0.5	0.5	0.455	nan	0.833	0.5	0
lr	wrap	none	0.833	0.5	0.5	0.455	nan	0.833	0.5	0
dummy	embed_lgbm	lgbm	0.833	0.5	0.5	0.455	nan	0.833	0.5	0

knn	wrap	none	0.833	0.5	0.5	0.455	nan	0.833	0.5	0
dummy	wrap	none	0.833	0.5	0.5	0.455	nan	0.833	0.5	0
dummy	pred	none	0.833	0.5	0.5	0.455	nan	0.833	0.5	0
dummy	none	none	0.833	0.5	0.5	0.455	nan	0.833	0.5	0
dummy	embed_linear	linear	0.833	0.5	0.5	0.455	nan	0.833	0.5	0
sgd	wrap	none	0.833	0.57	0.519	0.494	0.429	0.839	0.519	0.046
rf	wrap	none	0.832	0.539	0.545	0.546	0.543	0.846	0.545	0.114
lgbm	wrap	none	0.829	0.552	0.514	0.489	0.378	0.837	0.514	0.043

2. Training 70% Validation 10% and Testing 20%

Number of Epochs: 10

Learning Rate: 0.001

Test Accuracy: 81.5%

Results:

Holdout Set Performance

The performance of the models using ResNet-50 embeddings with a 70-10-20 data split shows strong results for most of the algorithms. The Light Gradient Boosting Machine (LGBM) with embedding-based feature selection (LGBM embeddings) achieved the highest accuracy of 97.7%, along with an AUROC of 98.8% and balanced accuracy of 94.8%. Logistic Regression (LR) with similar feature selection also performed well, achieving an accuracy of 97.5%, AUROC of 98.7%, and balanced accuracy of 94.1%.

K-Nearest Neighbours (KNN) performed comparably, with accuracy ranging from 97.3% to 97.5%, balanced accuracy around 92.6%, and an AUROC above 97.0% for most configurations. Stochastic Gradient Descent (SGD) models achieved accuracies around 96.9%, with balanced accuracies near 92.9%. Random Forest (RF) models demonstrated slightly lower accuracies, around 95.9%, and balanced accuracies of approximately 92.5%.

Simpler selection methods, such as "wrap," resulted in significantly lower accuracies and balanced accuracies, showing the importance of using advanced embedding-based feature selection strategies.

5-Fold Cross-Validation Performance on Holdout Set

The 5-fold cross-validation results align closely with the holdout set performance:

LGBM models with embedding-based selection led with an accuracy of 97.8%, AUROC of 98.8%, and balanced accuracy of 94.9%. LR models maintained high accuracy and balanced accuracy scores, with metrics similar to their holdout set performance. KNN models exhibited strong performance, with accuracies near 97.1% and balanced accuracies around 92.6%.

Holdout Set Performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lgbm	embed_lgbm	lgbm	0.977	0.988	0.948	0.957	0.953	0.981	0.948	0.906
lgbm	embed_linear	linear	0.975	0.987	0.941	0.954	0.96	0.978	0.941	0.889
knn	pred	none	0.975	0.973	0.934	0.954	0.977	0.975	0.934	0.873
lgbm	assoc	none	0.975	0.987	0.941	0.954	0.96	0.978	0.941	0.889
lgbm	none	none	0.974	0.987	0.94	0.952	0.952	0.978	0.94	0.889
knn	assoc	none	0.973	0.971	0.926	0.948	0.977	0.972	0.926	0.857
knn	embed_lgbm	lgbm	0.973	0.968	0.926	0.948	0.977	0.972	0.926	0.857
knn	embed_linear	linear	0.973	0.971	0.926	0.948	0.977	0.972	0.926	0.857
knn	none	none	0.973	0.971	0.926	0.948	0.977	0.972	0.926	0.857
lgbm	pred	none	0.969	0.983	0.931	0.943	0.938	0.975	0.931	0.873
sgd	embed_lgbm	lgbm	0.969	0.984	0.929	0.942	0.938	0.974	0.929	0.869
rf	embed_linear	linear	0.963	0.975	0.927	0.932	0.899	0.975	0.927	0.873
rf	embed_lgbm	lgbm	0.959	0.975	0.925	0.926	0.88	0.975	0.925	0.873
rf	none	none	0.956	0.974	0.918	0.92	0.871	0.972	0.918	0.861
lr	embed_lgbm	lgbm	0.955	0.977	0.891	0.914	0.924	0.96	0.891	0.795
rf	assoc	none	0.951	0.975	0.908	0.911	0.858	0.969	0.908	0.844
rf	pred	none	0.941	0.963	0.881	0.891	0.846	0.959	0.881	0.791
sgd	assoc	none	0.941	0.894	0.894	0.893	0.82	0.965	0.894	0.824
sgd	none	none	0.937	0.883	0.883	0.885	0.813	0.961	0.883	0.803
sgd	embed_linear	linear	0.935	0.877	0.877	0.881	0.811	0.959	0.877	0.791
sgd	pred	none	0.929	0.877	0.865	0.869	0.793	0.955	0.865	0.77
lr	embed_linear	linear	0.919	0.924	0.764	0.82	0.963	0.915	0.764	0.533
lr	none	none	0.919	0.924	0.764	0.82	0.963	0.915	0.764	0.533
lr	assoc	none	0.919	0.924	0.764	0.82	0.963	0.915	0.764	0.533
knn	wrap	none	0.918	0.866	0.805	0.835	0.829	0.931	0.805	0.635
rf	wrap	none	0.917	0.92	0.798	0.832	0.839	0.928	0.798	0.619
lr	pred	none	0.916	0.925	0.751	0.808	0.976	0.91	0.751	0.504
lgbm	wrap	none	0.914	0.919	0.801	0.83	0.811	0.93	0.801	0.631
sgd	wrap	none	0.914	0.878	0.759	0.81	0.915	0.913	0.759	0.529
lr	wrap	none	0.913	0.914	0.759	0.809	0.908	0.913	0.759	0.529
dummy	embed_lgbm	lgbm	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	wrap	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	pred	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	none	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	embed_linear	linear	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	assoc	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0

5 – fold performance on Holdout Set

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lgbm	embed_lgbm	lgbm	0.978	0.988	0.949	0.96	0.962	0.982	0.949	0.906
lgbm	embed_linear	linear	0.976	0.989	0.945	0.956	0.959	0.98	0.945	0.897
lgbm	assoc	none	0.975	0.988	0.941	0.954	0.962	0.978	0.941	0.889
lgbm	none	none	0.975	0.989	0.937	0.952	0.966	0.977	0.937	0.881
sgd	embed_lgbm	lgbm	0.972	0.985	0.937	0.948	0.946	0.978	0.937	0.885
knn	embed_linear	linear	0.971	0.968	0.919	0.944	0.986	0.969	0.919	0.84
knn	pred	none	0.971	0.969	0.924	0.945	0.973	0.972	0.924	0.852
knn	embed_lgbm	lgbm	0.971	0.971	0.922	0.943	0.974	0.971	0.922	0.848
knn	assoc	none	0.97	0.968	0.916	0.942	0.982	0.969	0.916	0.836
knn	none	none	0.97	0.968	0.916	0.942	0.982	0.969	0.916	0.836
lgbm	pred	none	0.965	0.985	0.922	0.935	0.934	0.972	0.922	0.856
rf	embed_lgbm	lgbm	0.956	0.977	0.919	0.92	0.87	0.973	0.919	0.865
rf	embed_linear	linear	0.953	0.98	0.918	0.915	0.856	0.973	0.918	0.864
lr	embed_lgbm	lgbm	0.951	0.974	0.87	0.902	0.952	0.953	0.87	0.75
rf	none	none	0.949	0.971	0.902	0.906	0.858	0.967	0.902	0.832
rf	assoc	none	0.944	0.976	0.9	0.898	0.826	0.967	0.9	0.836
rf	pred	none	0.94	0.961	0.869	0.886	0.861	0.954	0.869	0.762
sgd	embed_linear	linear	0.937	0.882	0.882	0.886	0.828	0.96	0.882	0.799
sgd	assoc	none	0.937	0.882	0.882	0.886	0.826	0.96	0.882	0.799
sgd	none	none	0.936	0.88	0.88	0.884	0.822	0.96	0.88	0.795
sgd	pred	none	0.926	0.879	0.864	0.866	0.788	0.955	0.864	0.77
knn	wrap	none	0.918	0.862	0.782	0.823	0.893	0.922	0.782	0.578
lr	none	none	0.918	0.907	0.755	0.809	0.988	0.912	0.755	0.512
lr	assoc	none	0.918	0.907	0.755	0.809	0.988	0.912	0.755	0.512
lr	embed_linear	linear	0.917	0.907	0.753	0.807	0.988	0.911	0.753	0.508
sgd	wrap	none	0.915	0.873	0.757	0.807	0.945	0.913	0.757	0.52
lgbm	wrap	none	0.914	0.924	0.784	0.82	0.855	0.924	0.784	0.59
lr	wrap	none	0.914	0.914	0.755	0.805	0.947	0.912	0.755	0.516
lr	pred	none	0.914	0.905	0.745	0.799	0.987	0.908	0.745	0.491
rf	wrap	none	0.908	0.923	0.787	0.814	0.797	0.926	0.787	0.606
dummy	embed_lgbm	lgbm	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	wrap	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	pred	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	none	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	embed_linear	linear	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	assoc	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0

3. Training 40% Validation 30% and Testing 30%

Number of Epochs: 10

Learning Rate: 0.001

Test Accuracy: 81.5%

Results:

Holdout Set Performance

The results for the ResNet-50 embeddings with a 40-30-30 data split demonstrate that the models performed well overall. Light Gradient Boosting Machine (LGBM) with embedding-based feature selection (LGBM embeddings) achieved an accuracy of 96.7%, an AUROC of 98.2%, and a balanced accuracy of 90.2%. Logistic Regression (LR) with LGBM embeddings followed closely, with an accuracy of 96.4%, AUROC of 98.5%, and balanced accuracy of 90.7%.

K-Nearest Neighbors (KNN) models also showed strong results, with accuracies ranging from 96.3% to 96.7% and balanced accuracies around 89.4% to 90.5%. Random Forest (RF) and Stochastic Gradient Descent (SGD) models performed slightly lower, with accuracies between 95.1% and 95.7% and balanced accuracies ranging from 88.0% to 91.5%, depending on the feature selection strategy.

5-Fold Cross-Validation Performance on Holdout Set

The 5-fold cross-validation results are consistent with the holdout set findings:

LGBM with LGBM embeddings achieved an accuracy of 95.7%, an AUROC of 97.6%, and a balanced accuracy of 89.6%. LR and KNN models maintained high performance, with accuracies around 95.4% to 95.9% and balanced accuracies exceeding 88.5%. RF and SGD models showed slightly lower performance compared to LGBM, with accuracies ranging from 94.0% to 94.9% and balanced accuracies around 88.0%.

Holdout Set Performance

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lgbm	embed_lgbm	lgbm	0.967	0.982	0.902	0.935	0.991	0.963	0.902	0.806
knn	pred	none	0.967	0.965	0.905	0.935	0.983	0.964	0.905	0.813
lr	embed_lgbm	lgbm	0.964	0.985	0.907	0.931	0.958	0.965	0.907	0.82
lgbm	none	none	0.964	0.983	0.907	0.931	0.958	0.965	0.907	0.82
knn	assoc	none	0.963	0.969	0.894	0.927	0.982	0.96	0.894	0.791
knn	embed_linear	linear	0.963	0.969	0.894	0.927	0.982	0.96	0.894	0.791
knn	none	none	0.963	0.969	0.894	0.927	0.982	0.96	0.894	0.791
lgbm	assoc	none	0.962	0.983	0.894	0.925	0.973	0.96	0.894	0.791
lgbm	pred	none	0.962	0.985	0.899	0.926	0.957	0.963	0.899	0.806
lgbm	embed_linear	linear	0.959	0.985	0.892	0.921	0.957	0.96	0.892	0.791
lr	none	none	0.957	0.98	0.885	0.916	0.956	0.957	0.885	0.777
lr	embed_linear	linear	0.957	0.98	0.885	0.916	0.956	0.957	0.885	0.777
lr	assoc	none	0.957	0.98	0.885	0.916	0.956	0.957	0.885	0.777
knn	embed_lgbm	lgbm	0.957	0.965	0.885	0.916	0.956	0.957	0.885	0.777
lr	pred	none	0.956	0.981	0.881	0.913	0.955	0.956	0.881	0.77
rf	pred	none	0.955	0.981	0.889	0.913	0.924	0.96	0.889	0.791
rf	embed_linear	linear	0.955	0.978	0.895	0.914	0.911	0.962	0.895	0.806
rf	embed_lgbm	lgbm	0.951	0.98	0.89	0.908	0.895	0.961	0.89	0.799
sgd	assoc	none	0.949	0.9	0.9	0.906	0.858	0.966	0.9	0.827
rf	none	none	0.949	0.974	0.88	0.902	0.9	0.957	0.88	0.777
sgd	embed_lgbm	lgbm	0.949	0.978	0.915	0.909	0.833	0.973	0.915	0.863
rf	assoc	none	0.948	0.977	0.882	0.9	0.886	0.958	0.882	0.784
sgd	pred	none	0.937	0.873	0.873	0.883	0.831	0.956	0.873	0.777
sgd	embed_linear	linear	0.933	0.93	0.876	0.879	0.803	0.959	0.876	0.791
sgd	wrap	none	0.931	0.931	0.812	0.856	0.926	0.931	0.812	0.633
lr	wrap	none	0.928	0.944	0.793	0.845	0.965	0.924	0.793	0.59
lgbm	wrap	none	0.926	0.95	0.812	0.849	0.881	0.932	0.812	0.64
knn	wrap	none	0.924	0.901	0.81	0.845	0.864	0.932	0.81	0.64
rf	wrap	none	0.915	0.804	0.759	0.812	0.936	0.913	0.759	0.525
sgd	none	none	0.909	0.882	0.833	0.835	0.73	0.944	0.833	0.719
dummy	embed_lgbm	lgbm	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	wrap	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	pred	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	none	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	embed_linear	linear	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	assoc	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0

5 – fold performance on Holdout Set

model	selection	embed_selector	acc	auroc	bal-acc	f1	npv	ppv	sens	spec
lr	embed_lgbm	lgbm	0.967	0.986	0.911	0.935	0.97	0.967	0.911	0.827
knn	embed_lgbm	lgbm	0.963	0.954	0.903	0.928	0.958	0.964	0.903	0.812
knn	pred	none	0.959	0.961	0.895	0.92	0.955	0.962	0.895	0.798
lgbm	assoc	none	0.959	0.982	0.895	0.921	0.953	0.962	0.895	0.798
lgbm	none	none	0.959	0.979	0.898	0.92	0.946	0.963	0.898	0.805
lgbm	pred	none	0.958	0.98	0.9	0.92	0.935	0.964	0.9	0.812
lgbm	embed_lgbm	lgbm	0.957	0.976	0.896	0.916	0.935	0.963	0.896	0.805
lgbm	embed_linear	linear	0.956	0.98	0.89	0.914	0.941	0.96	0.89	0.791
knn	assoc	none	0.954	0.957	0.886	0.909	0.925	0.959	0.886	0.784
knn	embed_linear	linear	0.954	0.957	0.886	0.909	0.925	0.959	0.886	0.784
knn	none	none	0.954	0.957	0.886	0.909	0.925	0.959	0.886	0.784
lr	assoc	none	0.949	0.979	0.868	0.898	0.939	0.952	0.868	0.748
lr	none	none	0.949	0.979	0.868	0.898	0.939	0.952	0.868	0.748
lr	embed_linear	linear	0.949	0.979	0.868	0.898	0.939	0.952	0.868	0.748
sgd	embed_lgbm	lgbm	0.948	0.975	0.908	0.905	0.84	0.97	0.908	0.848
lr	pred	none	0.946	0.98	0.864	0.894	0.932	0.951	0.864	0.741
rf	pred	none	0.94	0.974	0.883	0.891	0.855	0.961	0.883	0.798
rf	assoc	none	0.939	0.973	0.88	0.887	0.84	0.959	0.88	0.791
rf	embed_lgbm	lgbm	0.934	0.973	0.865	0.877	0.836	0.954	0.865	0.762
rf	none	none	0.926	0.968	0.861	0.865	0.793	0.953	0.861	0.762
sgd	wrap	none	0.925	0.918	0.823	0.852	0.85	0.937	0.823	0.67
rf	embed_linear	linear	0.925	0.969	0.857	0.862	0.793	0.952	0.857	0.755
lr	wrap	none	0.923	0.943	0.801	0.84	0.883	0.929	0.801	0.619
knn	wrap	none	0.92	0.882	0.805	0.838	0.85	0.931	0.805	0.634
rf	wrap	none	0.917	0.84	0.783	0.825	0.874	0.922	0.783	0.583
sgd	embed_linear	linear	0.912	0.894	0.84	0.841	0.742	0.947	0.84	0.734
lgbm	wrap	none	0.911	0.937	0.791	0.821	0.804	0.926	0.791	0.612
sgd	pred	none	0.907	0.832	0.832	0.833	0.727	0.944	0.832	0.719
sgd	assoc	none	0.906	0.831	0.831	0.831	0.722	0.944	0.831	0.719
sgd	none	none	0.903	0.854	0.827	0.827	0.715	0.943	0.827	0.712
dummy	embed_lgbm	lgbm	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	wrap	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	pred	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	none	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	embed_linear	linear	0.834	0.5	0.5	0.455	nan	0.834	0.5	0
dummy	assoc	none	0.834	0.5	0.5	0.455	nan	0.834	0.5	0

Comparison of Results Across Different Dataset Splits

Metric	80%-10%-10%	70%-10%-20%	40%-30%-30%
Logistic Regression (LR)	Accuracy: 98.3%	Accuracy: 97.5%	Accuracy: 96.4%
	Balanced Accuracy: 96.3%	Balanced Accuracy: 94.1%	Balanced Accuracy: 90.7%
	AUROC: 99.7%	AUROC: 98.7%	AUROC: 98.5%
Light Gradient Boosting Machine (LGBM)	Accuracy: 98.1%	Accuracy: 97.7%	Accuracy: 96.7%
	Balanced Accuracy: 95.7%	Balanced Accuracy: 94.8%	Balanced Accuracy: 90.2%
	AUROC: 99.7%	AUROC: 98.8%	AUROC: 98.2%
K-Nearest Neighbors (KNN)	Accuracy: 97.3%	Accuracy: 96.7%	Accuracy: 96.3%
	Balanced Accuracy: 92.4%	Balanced Accuracy: 90.5%	Balanced Accuracy: 89.4%
	AUROC: 98.2%	AUROC: 97.3%	AUROC: 96.9%
Stochastic Gradient Descent (SGD)	Accuracy: 97.3%	Accuracy: 96.9%	Accuracy: 96.4%
	Balanced Accuracy: 95.5%	Balanced Accuracy: 92.9%	Balanced Accuracy: 90.7%
	AUROC: 95.8%	AUROC: 94.6%	AUROC: 93.9%

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